

Engineering What an Optical Sensor Cannot See

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July 24, 2026

Abstract. A symmetry of an optical measurement law makes a scene subspace exactly invisible. We turn that blindness into a calibrated, certifiable design variable. A statistical Noether theorem identifies each exact wall with a score-null orbit; a blindness-capacity theorem, $\beta_d = \sigma_d$, makes the largest guaranteed-invisible code space a measured singular spectrum; a three-ledger theorem separates what finite data can certify into a $1 : \sqrt{p} : p$ ladder for known, blind, and estimated change; and a jet-transmutation corollary shows an exact wall or projection restores the higher-amplitude law. A scaling gate and a wave-optics twin validate the program. This opens calibrated, publishable optical blindness.

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1 Introduction

A strongly scattering or bandlimited optical channel is usually described by what it destroys. The received wisdom is that a lost coordinate is simply lost. The companion Letter [1] already separates two invariants that this view conflates: reconstruction rank, which can collapse to one, and local change observability, which can stay complete because the first nonzero nuisance-profiled statistical jet fixes the price of observing each direction. That Letter owns the Rank–Jet Separation theorem and the algebraic mean wall of a scrambling channel.

Here we take the complementary and constructive step. Rather than accept an optical sensor’s blindness as a defect, we ask what it *cannot* see, whether that ignorance can be measured, engineered, and certified, and what finite data then permit us to conclude. The result is a hierarchy of three theorems and one corollary that turns invisibility into a metrological primitive:

1. exact measurement symmetries create *walls*—score-null orbits on which no estimator, exposure count, or reconstruction prior can respond (Section 2);
2. calibration measures a wall’s *blindness capacity*—the largest code subspace guaranteed below a declared leak—as a singular spectrum, with a finite metrological envelope and an explicit noncomposability law (Section 3);
3. geometry and projection *select the jet order*, restoring the desired higher-amplitude channel through three physically distinct routes (Section 4);
4. unknown tangent dimension imposes a separate $1 : \sqrt{p} : p$ *certifiability ladder* on known, blind, and estimated change (Section 5).

The abstract ingredients—gauge nulls, singular-value min–max, and local asymptotic normality—are individually classical [7, 8]; the contribution is their synthesis into

a physical optical metrology program with a measured dimension–leak curve, a certified cannot-see guarantee, and a wave-optics transfer chain (Section 6). The five sections are organized by theorem, not by chronology; the seventeen-test verification battery, the honest negatives, and the full proofs are in the Supplement. Every quoted number traces to a single committed artifact.

2 Symmetry walls of an optical measurement

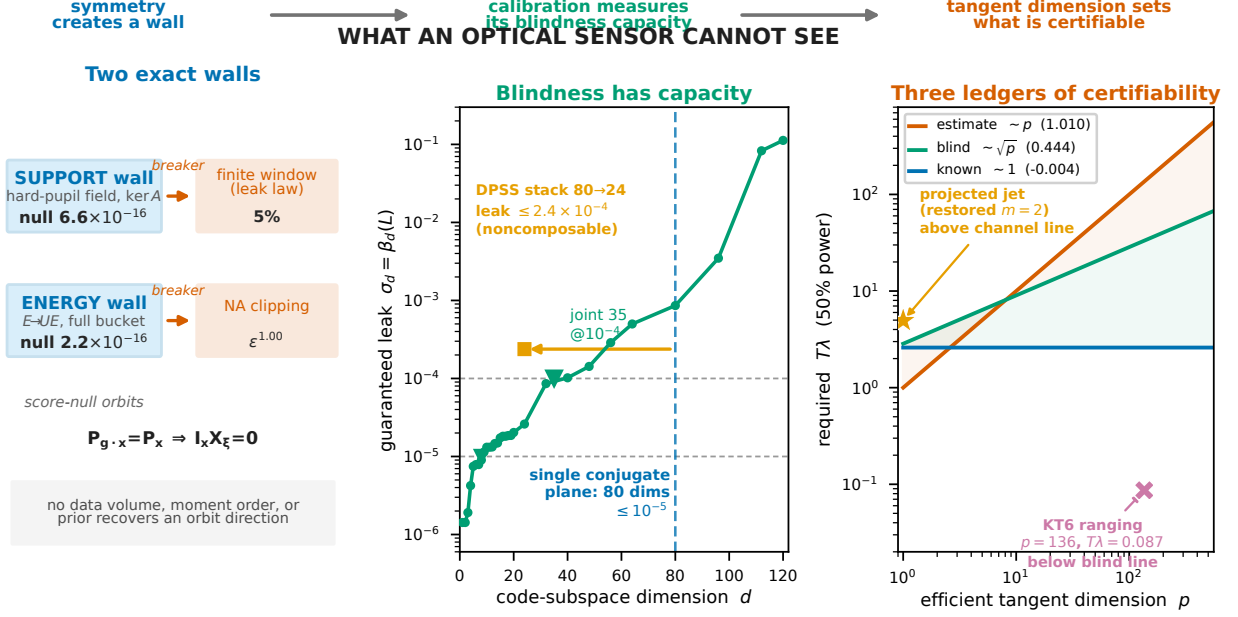
Let a scene x lie on a smooth parameter manifold, let the complete observed record have a dominated law P_x with density $p(y \mid x)$, and let a Lie group G act on the scene with infinitesimal generator X_ξ for a Lie-algebra element ξ .

Theorem 1 (Statistical Noether wall). *If the measurement law is G -invariant, $P_{g \cdot x} = P_x$ for every $g \in G$, then every orbit tangent is an exact score-null direction,*

$$D_x \log p(Y \mid x)[X_\xi(x)] = 0 \text{ a.s.}, \quad I_x X_\xi(x) = 0, \quad (1)$$

so every f -divergence, Chernoff distance, and quotient information jet between two points of one orbit vanishes: no estimator, moment order, exposure count, or reconstruction prior can distinguish them. Conversely, a smooth constant-rank integrable distribution that is score-null on a connected neighborhood has integral leaves that are local blindness orbits.

Proof sketch. Differentiate $\log p(y \mid \exp(t\xi) \cdot x) = \log p(y \mid x)$ at $t = 0$ to get (1); the score Gramian gives $I_x X_\xi = 0$, and equality of the complete laws gives zero divergence at every finite orbit displacement. The converse is Frobenius integrability applied to the score-null distribution. Full statement and the discrete-degeneracy caveat are in Supplement S1. $\square$



$\beta_d(L) = \sigma_d(L)$ — and does not automatically compose $1 : \sqrt{p} : p$ — a nonzero signal is not a certifiable signal

Figure 1. What the sensor cannot see. A left-to-right causal spine over three quantitative panels. *Left* — symmetry walls: two exact score-null orbits, the hard-pupil support (field) wall at a maximum leak beyond $2kR$ of 6.6×10^{-16} and the lossless full-aperture energy (unitary) wall at 2.18×10^{-16} (both machine zero), each with its one controlled breaker (finite-window leak law; ϵ^1 NA clipping). *Center* — blindness capacity: the generalized leak singular value σ_d versus code-subspace dimension d (single deterministic joint spectrum), with the single-plane point (80 dimensions $\leq 10^{-5}$ at the fixed conjugate plane), the joint spectrum point (35 dimensions $\leq 10^{-4}$ over four z_1 planes), and the hardware DPSS route ($\leq 2.4 \times 10^{-4}$, whose single-plane SVD capacity does not compose, $80 \rightarrow 24$). *Right* — certifiability ladder: the required $T\lambda \sim 1 : \sqrt{p} : p$ thresholds for known, blind, and estimated change, with the TLSG-measured p -exponents $-0.004/0.444/1.010$; the near-contact KT6 operating point sits below the blind line, and the projected $m = 2$ jet sits above its known-direction channel line. All values are committed frozen artifacts (KT1 @19338a4, IT1 @f13a323, IT4b @5666d1e, IT5 @c8a0487, KT6 @635ff05, TLSG @35d858e).

The honest content of Theorem 1 is that a symmetry of the *wave equation* does not by itself create a wall; a wall appears only when the *full measurement law* inherits the induced parameter action. Two walls of distinct physical type realize it exactly.

Support (field) wall. For a bandlimited mean operator A , every $h \in \ker A$ generates a translation $x \mapsto x + h$ that leaves the mean-channel law invariant—the additive group of the unmeasured subspace. A hard Fourier pupil restores this wall to machine precision: the maximum signed-intensity field leak beyond $2kR$ is 6.6×10^{-16} across all hard-pupil variants and all defocus z_1 , and the ideal periodic-code detector wall is 2.07×10^{-15} for $k > k_p$.

Energy (unitary) wall. A lossless full-aperture bucket measures only the Hilbert-space norm $B(E) = \|E\|_2^2$, so $B(UE) = B(E)$ for every unitary spatial transformation U . A phase-only object change is one orbit of this wall; the measured null is 2.18×10^{-16} at every order and for every (ϵ, z_2) . This is the diagonal subgroup of a much larger *mode-unitary* wall: the energy bucket is blind not only to

phase screens but to any unitary spatial-mode transformation.

The two controlled breakers. Each wall has exactly one designed breaker. NA clipping replaces the identity collector by a projector P_{NA} ; $\|P_{NA}UE\|^2$ is no longer invariant and the energy wall breaks at first order, with clip-bucket slopes 0.97–1.03 that are monotone in z_2 . A finite object window breaks the support wall: the full twin develops a noncompact pedestal with detector leak beyond $2k_p$ equal to 5.3×10^{-2} (field leak 1.6×10^{-1}), so frequency placement alone gives no wall—an exact detector wall additionally requires a concentration-grade guard (Section 6). These two group actions and their breakers are the first entries of a wall periodic table for an optical chain (Supplement S1); a parity-symmetric chain and a polarization-insensitive bucket supply two further rows.

3 Measuring blindness capacity

An exact wall is a limiting case. In practice one wants the *largest* code space on which a sensor is guaranteed not to

respond to a declared nuisance, and a number certifying it. Let a calibrated linear leak operator $L : \mathcal{C} \rightarrow \mathcal{Y}$ map a code to its forbidden mean-channel response, with the code norm whitened by the desired in-band metric, and write its singular values ascending, $\sigma_1 \leq \dots \leq \sigma_n$. Define the best worst-case leak of a d -dimensional code subspace,

$$\beta_d(L) = \min_{\dim S=d} \max_{c \in S, \|c\|=1} \|Lc\|. \quad (2)$$

Theorem 2 (Blindness capacity). $\beta_d(L) = \sigma_d(L)$, attained by the span of the bottom d right singular vectors; no other d -dimensional code family guarantees a smaller worst-case leak. For a generalized relative-leak certificate with desired-response metric $B^\dagger B$, $\beta_d = \sqrt{\lambda_d}$ where $L^\dagger L v = \lambda B^\dagger B v$. If a calibration returns $\hat{L}$ with $\|L - \hat{L}\|_{\text{op}} \leq \delta$, the bottom- d subspace of $\hat{L}$ obeys the metrological envelope

$$\max_{c \in S_d, \|c\|=1} \|Lc\| \leq \sigma_d(\hat{L}) + \delta. \quad (3)$$

Proof sketch. Equation (2) is the Courant–Fischer min–max characterization of σ_d : the bottom- d subspace attains σ_d , and any d -dimensional subspace meets the orthogonal complement of the bottom $d - 1$ singular vectors, so it contains a unit vector with leak at least σ_d . The generalized form is the same statement for the pencil $(L^\dagger L, B^\dagger B)$; (3) is Weyl’s inequality. Supplement S2. $\square$

Theorem 2 makes “cannot see” a measured spectrum. At the fixed relay-conjugate plane, 80 code dimensions are certified below 10^{-5} (and below 10^{-4})—a single-plane blindness-capacity point, not an incidental code-design success. Enforcing invisibility *jointly* over four defocus planes $z_1 \in \{0, 2, 5, 10\}$ mm costs capacity: 35 dimensions remain below 10^{-4} jointly (8 below 10^{-5}), so the single-plane requirement buys 45 extra dimensions. The full ascending spectrum runs from $\sigma_1 = 1.43 \times 10^{-6}$ through $\sigma_{64} = 4.99 \times 10^{-4}$ to $\sigma_{120} = 1.13 \times 10^{-1}$: the plotted capacity curve of Fig. 1. In the $d = 64$ subspace the worst-direction leak is 4.99×10^{-4} , while a random in-subspace code leaks 1.68×10^{-4} in RMS—threefold below the worst direction and ridge-stable (d at 10^{-4} is 35 across regularizers $0/10^{-8}/10^{-6}$). Certified-blind codes retain a second sensing channel: the random-to-blind eigenvalue ratio is only 1.20 ($\ll 3$), yet the operational mean leak is 219 $\times$ lower (2.58×10^{-4} versus 5.64×10^{-2}).

The envelope (3) is what upgrades a numerical null space to a certified instrument property. On a fresh calibration/test split the simultaneous $\sigma_{64}(\hat{L}) + \delta$ envelope has coverage 1.000 ($\sigma_{64} = 3.4 \times 10^{-8}$, $\delta = 1.07 \times 10^{-2}$, worst fresh bottom-64 leak 1.99×10^{-6} , six orders below the envelope), and at least 64 conjugate-plane dimensions stay certified below 10^{-4} on fresh physical-law draws, with a measured minimum of 80.

4 Restoring the desired jet

The exact walls of Section 2 are the endpoints of a continuous mechanism. Let the profiled Chernoff distance carry two leading operators, $C(\varepsilon, g) = A g^{2a_1} \varepsilon^{2m_1} + B g^{2a_2} \varepsilon^{2m_2}$ with $0 < m_1 < m_2$, and set $y = (B/A) g^{2(a_2-a_1)} \varepsilon^{2(m_2-m_1)}$. The effective amplitude order $m_{\text{eff}} = \frac{1}{2} d \log C / d \log \varepsilon = (m_1 + m_2 y) / (1 + y)$ obeys the exact flow

$$\frac{dm_{\text{eff}}}{d \log \varepsilon} = 2(m_{\text{eff}} - m_1)(m_2 - m_{\text{eff}}), \quad (4)$$

with fixed points m_1 and m_2 .

Corollary 1 (Jet transmutation). A lower-order symmetry-breaking channel ($m_1 = 1$) drives the strict local observability class to $m_{\text{eff}} \rightarrow 1$ as $\varepsilon \rightarrow 0$; setting its coefficient to zero by an exact wall, a certified code, or a nuisance projection places the experiment on the higher-order fixed manifold $m_2 = 2$ without changing the scene perturbation. In the near-contact optical case $m_1 = 1, a_1 = 0, m_2 = 2, a_2 = 2$, so the crossover invariant is $\varepsilon z_2^2 = \text{const}$, i.e. $z_{2,*} \propto \varepsilon^{-1/2}$.

Three physically distinct routes realize the corollary, and they are alternatives, not factors.

Route 1 — optical projection. Projecting out the measured mean-leak tangent restores the covariance-only slope to 4.00 in all four cells, against an unprojected leak-dominated slope of 2.12–2.20; the orthogonality residual is $\sim 10^{-18}$. A generic direction, by contrast, stays sub-quartic (slopes 1.58–2.28).

Route 2 — code-space projection. Projecting the covariance score off the leak/law span drives the nuisance overlaps to zero (amplitude $0.36 \rightarrow 0$, medium $0.20 \rightarrow 0$) while retaining 86–99% of the target information.

Route 3 — hardware concentration. A discrete prolate spheroidal (DPSS) [9] render-and-object window holds the beyond- $2k_p$ witness leak below 2.4×10^{-4} uniformly in z_1 (maximum 2.38×10^{-4}), 349 $\times$ below the no-window value 8.3×10^{-2} ; a naive separable Tukey window reaches only $\sim 4 \times 10^{-2}$ (rim-limited).

These routes do not stack. For several leak routes $L_1, \dots, L_k$ the simultaneous root-sum-square certificate is governed by the stacked Gram operator, so

$$\beta_d^2 = \lambda_d^\dagger \left(\sum_i w_i^2 L_i^\dagger L_i \right), \quad (5)$$

which is not multiplicative in the individual certificates. Concretely, the DPSS hardware route does not compose with the SVD code route: the single-plane certified dimension drops from 80 to 24 at 10^{-4} , because the window rotates and reshapes the operator and trades code-space capacity. A second, independent witness of the same law is the

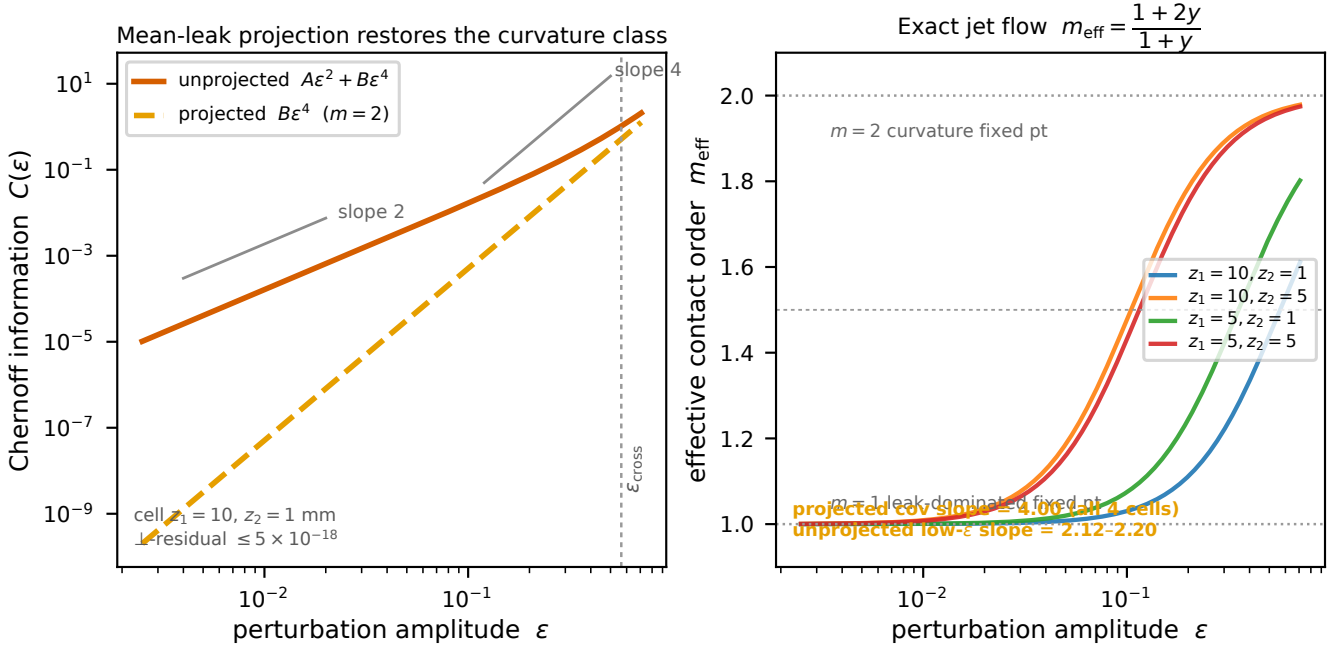


Figure 2. Jet transmutation. *Left:* the profiled covariance distance $C(\epsilon) = A\epsilon^2 + B\epsilon^4$ reconstructed from the committed fitted coefficients of the four KT2 cells. Unprojected, a first-order mean leak ($A > 0$) gives slope 2 ($m = 1$) at small amplitude; projecting the measured mean-leak tangent zeroes A and leaves slope 4 ($m = 2$) everywhere, with the crossover at $\epsilon_{\text{cross}} = \sqrt{A/B}$. *Right:* the exact logistic jet flow $dm_{\text{eff}}/d \log \epsilon = 2(m_{\text{eff}} - 1)(2 - m_{\text{eff}})$ between the integer fixed points 1 and 2; the four measured projected slopes are 4.00 and the unprojected low-amplitude slopes are 2.12–2.20. Cross-anchor: the companion integer contact orders 2.038/4.000 (Ref. [COMPANION]). Source: KT2 @05f1029.

Tukey apodization, whose joint certified dimension drops from 35 to 24 and which is $93\times$ worse at $d = 64$ (apodized relative leak 4.66×10^{-2} versus 4.99×10^{-4} ; a rim effect). Independent hardware modifications change the leak operator and its metric and must be recalibrated jointly.

5 What finite data can certify

A wall says a change is invisible; blindness capacity says how large a subspace is guaranteed invisible; the jet order says how a visible change scales. A fourth, independent price is set by finite data and unknown direction. Whiten the null covariance, profile every admitted nuisance tangent, and let the efficient covariance displacement be $A_{\perp}$ in a symmetric tangent space of dimension p , with efficient signal $\lambda = \frac{1}{2} \|A_{\perp}\|_F^2$. For T independent banks the aggregate score experiment is locally $Z_T \approx \mathcal{N}(\sqrt{T}\theta, I_p)$ with $\|\theta\|^2 = \lambda$.

Theorem 3 (Three-ledger certifiability). *Under this nuisance-profiled reduction, the per-task bank budget for fixed nontrivial power obeys*

$$T\lambda \sim \begin{cases} 1, & \text{known direction (matched score),} \\ \sqrt{p}, & \text{blind existence (unknown direction),} \\ p, & \text{estimate or localize the direction,} \end{cases} \quad (6)$$

a $1 : \sqrt{p} : p$ ladder. Composed with a quotient jet $A_{\perp} = \kappa g^a \epsilon^m B$, the known-direction budget is $T_{\text{known}} \propto$

$\kappa^{-2} g^{-2a} \epsilon^{-2m}$, and the blind and estimation budgets multiply it by $\sqrt{p}$ and p ; adaptation changes the prefactor, never the physical exponent $2m$.

Proof sketch. The matched score $u^{\top} Z_T$ separates the hypotheses at $T\lambda = O(1)$. For the blind ledger the invariant $\|Z_T\|^2$ is χ_p^2 versus $\chi_p^2(T\lambda)$, separated when $T\lambda \gg \sqrt{p}$; a uniform spherical mixture gives a matching minimax lower bound, since its χ^2 -divergence from the null scales as $T^2 \lambda^2 / p$ and vanishes below $T\lambda \sim \sqrt{p}$. The efficient estimator $\hat{\theta} = Z_T / \sqrt{T}$ has risk p/T , matching the Cramér–Rao bound, so relative error is controlled only at $T\lambda \sim p$. Supplement S4. $\square$

A dedicated scaling gate (TLSG) confirms the ladder without a repair round. The fitted p -exponents for the known, blind, and estimation tasks are $-0.004/0.444/1.010$ against the frozen targets $0, \frac{1}{2}, 1$. The matched-score 50%-power contours collapse against $T\lambda$ with coefficient of variation $\text{CV} = 0.125$; the blind contours collapse against $T\lambda/\sqrt{p}$ ($\text{CV} = 0.124$) and fail against $T\lambda$ alone ($\text{CV} = 0.628$), the signature of the $\sqrt{p}$ adaptation penalty; the estimation risk collapses against $T\lambda/p$ ($\text{CV} = 0.035$, mean risk $\cdot T\lambda/p = 0.985$, i.e. the Cramér–Rao floor p/T).

The ladder also predicts a negative for the right reason. The near-contact channel-ratio operating point (KT6) sits below the optimal blind tier: its blind detection power is 0.051 at a false-positive rate of 0.05 (rising to 0.056 and 0.134 at $10\times$ and $100\times$ exposure), with $p = 136$ and $T\lambda =$

0.0866. The oracle law is nonetheless intact—the near-contact ratio $J_{\text{cov}}/J_{\text{mean}} \propto z_2^4$ with fitted exponents 3.99–4.08—but the blind signal-to-floor ratio is $5.4 \times 10^{-3} \ll 1$, below the Wishart M^2/T floor [10], so the estimator is killed. This is a boundary of the third ledger, not a failure of apparatus: the projected $m = 2$ covariance channel of Section 4 (slope 4.00) is by contrast a *known*-direction observable, and sits above its channel line in Fig. 1.

6 Wave-optics transfer and apparatus constraints

The theorems above are algebraic; a bench is not. On the unfiltered complementary-DMD twin the exact mean wall becomes a reproducible calibrated residual,

$$L(z_1) = 6.94 \times 10^{-3} + 5.92 \times 10^{-3} (z_1/\text{mm})^{0.90}, \quad (7)$$

whose floor, coefficient, and exponent are stable across code seeds and code counts with coefficients of variation 2.2%, 2.2%, and 0.1%. The conjugate-plane floor is an instrument constant set by fill-factor and macro-pixel staircase effects (moving 6% and 50% respectively, with $L_{\text{pix}} = 7.15 \times 10^{-3}$); the pupil does not reduce it. Deterministic beyond-band leak grows with defocus, from 6.8×10^{-3} at the conjugate plane to 5.4×10^{-2} at $z_1 = 10$ mm and 8.9×10^{-2} at $z_1 = 20$ mm; near-conjugate relay imaging suppresses the propagation term. A full-Maxwell check anchors the model: the thin-phase-screen speckle contrast 0.467 agrees with a two-dimensional finite-element (COMSOL 6.3 Wave Optics, TE) contrast 0.395 to 15.5%. The covariance signal grows as $\lambda \propto M^{1.8-1.9}$ in code count (fitted exponent 1.78 at mid range, 1.89 near contact), and the geometry-matched developed cell extrapolates to ≈ 768 code banks at 2% error for $M = 128$, inside the sealed band [513, 2996].

Two apparatus rules follow. First, because a separable Tukey guard reduces the detector tail only $\sim 8\text{--}40\times$ and never to 10^{-12} , an exact physical detector wall requires a concentration-grade (DPSS) guard together with the hard Fourier pupil—the choice of Section 4, Route 3. Second, binary rendering must use full-period pulse-width modulation with whole-period bucket integration: Bayer-ordered spatial dithering doubles the conjugate-plane leak, from 7.0×10^{-3} to 1.38×10^{-2} , above the 10^{-2} tolerance.

7 Discussion

Blindness with a symmetry origin is not a defect to be hidden but a quantity to be calibrated, engineered, and certified. The four objects are independent: a symmetry decides whether a change exists in the record, the jet order decides how its signal scales, the singular spectrum decides

how large a code space is guaranteed blind, and the tangent dimension decides whether that signal can be blindly certified. A physical sensor can therefore *publish* a calibrated dimension–leak frontier describing the largest code space on which it is guaranteed not to respond to a declared nuisance, backed by the metrological envelope (3) rather than an uncertified numerical null. The closest structural cousins—decoherence-free subspaces [3, 4], dynamical decoupling [5], and unobservable subspaces in nonlinear control [6]—share the kernel idea; the optical novelty is the hardware-calibrated, task-selective blindness with a measured capacity curve and a finite-sample certifiability ladder. The program is optics-first by design: without a bench instrument or a second non-optical physical realization the synthesis belongs in Optica, and the memory-effect phase diagram [2] (which requires a genuine thickness or stacking axis) is deferred. The same symmetry-first accounting extends to quantum-limited sensing, where a measurement symmetry likewise sets what no estimator can resolve [11]. The honest negatives that bound the scope—a segmented bucket that breaks the mean wall, and a coincidence veto that repairs medium false alarms while retaining only 24% of scene power—appear with the positives in Supplement S5.

Back matter

Funding. [FUNDING/ACKNOWLEDGMENTS]

Acknowledgments. [FUNDING/ACKNOWLEDGMENTS]

Disclosures. The authors declare no conflicts of interest.

Data availability. The committed frozen artifacts underlying every numerical claim, the claim–source matrix, and the figure scripts are available at [URL]. See Supplement 1 for the full verification battery, proofs, and the TLSG gate table.

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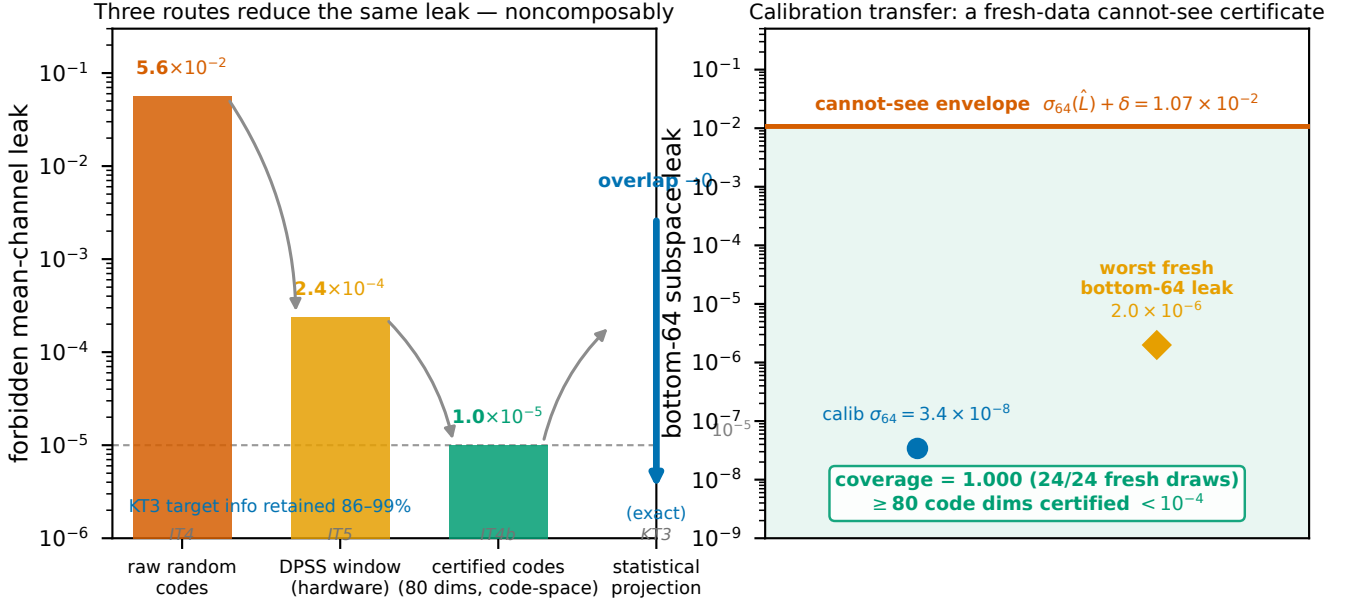


Figure 3. Restoration routes and calibration transfer. *Left:* the descending leak ladder of the three physically distinct routes plus the exact statistical projection, all reducing the forbidden mean-channel response—raw random codes 5.6×10^{-2} , DPSS window 2.4×10^{-4} , certified codes $\leq 10^{-5}$ (single-plane, 80 dimensions), statistical projection to 0 (exact)—which are alternatives, not factors. *Right:* calibration transfer (Eq. (3)) on a fresh calibration/test split: the simultaneous cannot-see envelope is 1.07×10^{-2} , the worst fresh bottom-64 leak is 2.0×10^{-6} (six orders below), coverage is 100% on 24 fresh draws, and at least 80 code dimensions remain certified below 10^{-4} at the conjugate plane. Sources: IT4 @33b8ef2, IT5 @c8a0487, IT4b @5666d1e, KT3 @2430166, TLSG @35d858e.

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